MILITARY SPECIFICATION
RELAY PANEL
RYK-13/A24B-4(V)

This specification has been approved by the
Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be used as a component in an Aircraft Monitor and Control (AMAC) System A/A24B-4(V). Relay Panel RYK-13/A24B-4(V) shall relay arming option and delivery mode signals to a nuclear weapon.

1.2 Classification - The equipment covered by this specification shall consist of one major item.

1.3 Associated Equipment - This equipment shall operate with the associated equipment listed in 6.5.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of

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MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for

STANDARDS

Military

MIL-STD-461	Electromagnetic Interference Requirements for Equipment
MIL-STD-704	Electric Power, Aircraft Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of

DRAWINGS

Naval Air Systems Command

DL 68A27E1	Data List for Relay Panel RYK-13/A24B-4(V) and all drawings therein

PUBLICATIONS

Military

ANA Bulletin No. 400	Electronic Equipment; Aircraft and Guided Missiles, Applicable Documents

2.1.1 Availability of Documents - When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications and documents required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials -

3.2.1 General Requirements - The parts and materials specified in the drawings and related documents for use in construction of this equipment were selected to ensure the fulfillment of all requirements of this specification. Any deviation by the contractor from the use of these specified parts and materials necessarily places both the burden of proof for the acceptability of the change, and the cost

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associated therewith, upon the contractor. Approval for deviations must be requested from the procuring activity by the contractor and granted by the procuring activity prior to submission of equipments for preproduction tests. Approval of parts and material deviations shall not be construed by the contractor to constitute a waiver of the performance requirements or any other requirements of this specification.

3.2.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained by the contractor as outlined in MIL-E-5400.

3.2.2 Supplemental Requirements - In the event that the contractor proposes to alter the equipment design delineated by this specification and applicable drawings, the following supplemental requirements for the selection of parts and materials are applicable.

3.2.2.1 Basic Conformance - Parts and materials selected for use in the equipment shall conform to the requirements of MIL-E-5400. Parts shall be selected from those parts included in ANA Bulletin No. 400, unless the parts listed therein will not perform adequately in the intended applications and environments specified herein.

3.2.2.2 Interchangeability - The equipment shall meet the interchangeability requirements of MIL-E-5400. The drawings listed in DL 68A27E1 shall be the standard for measuring interchangeability of all applicable units, assemblies, subassemblies, and parts.

3.3 Design and Construction - The equipment shall be manufactured in accordance with the drawings and subsidiary specifications listed in DL 68A27E1 and related lists, except insofar as errors of omission or commission on the drawings may be incompatible with the requirements of this and related specifications. In any case it is a condition of final acceptance that the equipment manufactured meet all the performance requirements of this specification, and such acceptance test documents as may be applicable. Proposed changes, either to correct drawing discrepancies or to effect improvements, shall be specifically approved by the procuring activity prior to their incorporation.

3.3.1 Total Weight - The total weight of the equipment, excluding cables, shall not exceed 4 pounds, 3 ounces.

3.3.2 Reliability -

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 50 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have a minimum total operating life of 20,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement due to wear during the life of the equipment, and the wearout life of such parts, shall be determined by the contractor and submitted to the procuring agency.

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3.3.2.3 Reliability in Mean-Time-Between-Failure (MTBF) - The equipment shall have a minimum of 2000 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.3.3 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be within the limits of MIL-STD-461, Class ID except as noted below:

Test Code	Extended Limits
CE01	Transient 50 db
CE02	Transient 50 db
CE03	Transient 40 db
CE04	Transient 40 db
RE02	Transient 20 db

3.3.4 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25 ± 5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative
Input power voltage	27.5 ± 0.5 volts d-c

3.3.5 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-E-5400 for class 2 equipment, except as modified herein.

3.3.5.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curve I of MIL-E-5400.

3.3.6 Warm-Up Time - The time required for the equipment to warm-up prior to operations shall not exceed 0.5 minute.

3.3.7 Input Electrical Power -

3.3.7.1 Operating Power - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when energized from the following power sources having characteristics and limits as defined in MIL-STD-704. The power required shall not exceed the specified amounts.

D-C Power, 28 volts, Category B, 30 VA

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3.3.7.2 Degraded Performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the equipment.

3.4 Performance - The performance of the equipment shall be as set forth below plus any and all performance requirements stated on the drawings of DL 68A27E1. Unless otherwise specified, these requirements shall apply under both standard and extreme service conditions. When reduced performance requirements for service conditions are acceptable, such variations are specified.

3.5 Detail Requirements - Relay Panel RYK-13/A24B-4(V) shall be manufactured in accordance with the documents listed on DL 68A27E1 and shall meet the requirements stated thereon and in the following subparagraphs.

3.5.1 Function - The relay panel shall relay arming option and delivery mode (ground-air burst, rate of descent, and fuze delay) signals from Control-Monitor DCK-143/A24B-4(V) and operating power from Transformer-Rectifier TEK-9/A24B-4(V) to a nuclear weapon, and weapon status signals from the weapon to the control-monitor.

3.5.2 Weight - The weight of the relay panel shall not exceed 4 pounds, 3 ounces.

3.5.3 Content - The relay panel shall contain the following components:

(1) Relays - Ten relays shall be provided to distribute the arming, safing, and delivery mode voltages to the weapon.

(2) Diode - A diode shall be provided in the safe circuit to prevent back voltages.

3.5.4 Electrical Characteristics - The electrical characteristics of the relay panel shall be as indicated in the following paragraphs.

3.5.4.1 Resistance Between Pins of J1 and Pins of J2 - The resistance between the pins of J1 and J2 shall be as specified in table 1.

Table 1. Resistance Between Pins of J1 and J2

From Pin of J1	To Pin of J2	Resistance in Ohms
G	G	Less than 0.01
N	R	Less than 0.01
C	C	Less than 0.01
J	W	Less than 0.01
U	V	Less than 0.01
D	F or E	Less than 0.01
E	F or E	Less than 0.01
F	b	830 ± 10%

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Table 1. Resistance Between Pins of J1 and J2 (Cont'd)

From Pin of J1	To Pin of J2	Resistance in Ohms
B	F or E	320 ± 10%
H	F or E	290 ± 10%
W	F or E	320 ± 10%
V	F or E	320 ± 10%
S	F or E	152 ± 10%
T	F or E	320 ± 10%
X	F or E	320 ± 10%

3.5.4.2 Continuity Between Pins of J1 and Pins of J2 - With pins D or E of J1 or pins E or F of J2 grounded, and with 27 ± 2 volts d-c applied to the pins of J1 specified in table 2, the following conditions shall exist:

(1) There shall be continuity (less than 0.05 ohm at 5 amperes) between the specified pins of J1 and J2.

(2) With 35 volts d-c applied to pins K and L of J1, current flow from pins K and L of J1 to pins A and P of J1 shall not exceed 1 milliampere when 27.5 volts d-c is applied to pins V and W of J1.

Table 2. Continuity Test

+27 ± 2 V D-C Applied to Following Pins of J1	Continuity Between Following Pins of J1 and J2
B	A (J1) and A (J2)*
B	P (J1) and A (J2)*
H and S	A or P (J1) and H, J, X, e and f (J2)
H, S and V	A or P (J1) and H, J, Y, X and f (J2)
H, S and T	A or P (J1) and H, J, e, d, and f (J2)
H, S and X	A or P (J1) and H, J, e, X and Z (J2)
W	K (J1) and A (J2)
W	L (J1) and A (J2)

*Resistance may be several hundred milliohms due to diode CR1 in series with the circuit.

3.5.4.3 Resistance Between Pins of J2 - Except for the resistance between pins F and E (which are unswitched returns), the resistance between each pin of J2 shall be no less than 100,000 ohms.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction (First Article) Tests

(2) Initial Production Tests

(3) Acceptance Tests

(4) Life Test

4.2 Preproduction (First Article) Tests - Preproduction tests shall be conducted by the contractor on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4.6. The Government Inspector and the procuring activity shall be advised when tests are to be conducted so that a government representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction (First Article) Test Data - The contractor shall submit all data collected in conducting these tests to the procuring activity for review.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification, other applicable specifications and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of MIL-T-5422 and interference tests in accordance with MIL-STD-461.

4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample.

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Evidence of non-compliance with the above shall constitute cause for rejection of an equipment and of equipment already accepted by the Government. It shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. No other tests shall be conducted on the equipment prior to starting the Initial Production Tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for initial production tests the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the initial production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the initial production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having adequate facilities for conducting all required tests shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Reliability Assurance Tests

(4) Special Tests

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4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run-In Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run-In Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment. This test shall be deleted if the reliability test includes a test on each equipment which consumes at least 6 hours of operation.

Temperature	Room ambient
Humidity	Room ambient
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6 hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in two directions the vertical direction shall be used. During the 6 hour period of operation following the 10 minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10 minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6 hour period specified above may be composed of two 3 hour periods to conform with standard working hours. This test is not required if a Reliability Assurance Test is selected which includes a test on each equipment which consumes at least 6 hours of operation.

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4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the Government Inspector in accordance with the following:

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	0 (See Note)
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

Sampling Tests are not required when Reliability Assurance Tests are conducted.

NOTE: If by contract action the initial production tests (4.3) are deleted, then a sample test shall be conducted on one equipment from the first ten produced.

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual, and special tests.

(3) Manufacturing run-in test specified in 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipments.

4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the Production Acceptance (Sampling) Phase shall be conducted.

4.4.3.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in table 5 of MIL-STD-781. For the qualification phase, Test Level F shall be used. The accept-reject criteria for Test Plan I shall be used.

4.4.3.2 Reliability Sampling Phase Tests - Samples of the equipment shall be tested as outlined in MIL-STD-781 under the section entitled "Sampling Phase of

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Production Reliability Tests". For the sampling phase, Test Level F shall be used. The accept-reject criteria for Test Plan I shall be used to determine the length of the tests (until an accept or reject decision is reached).

4.4.3.2.1 Lot Size for Sampling Phase - The equipments constructed during 1 month shall be one lot. The Accept-Reject criteria for Test Plan I shall be used to determine the length of the tests in each lot. Unless otherwise specified in the contract or order, no equipments shall be shipped until an Accept decision is reached under each lot.

4.4.3.3 Test Details - The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, and preventive maintenance to be allowed during the test, shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the qualification test phase of the reliability assurance tests.

4.4.4 Special Tests - Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Scope of Tests - Special Tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or a design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

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4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the individual tests. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the Government Inspector in accordance with the following. (Equipments which have successfully passed the initial production test, sampling tests, reliability tests, or special tests may be selected for life tests.) When reliability tests are conducted, the life test may be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0 - 5000 ft.)
Humidity	Room ambient
D-C Voltage	27.5 $\pm$ 2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

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(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the Government Inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification and the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished to the Government Inspector.

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5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555. In the event the equipment is not covered in MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6. NOTES

6.1 Intended Use - Relay Panel RYK-13/A24B-4(V) is intended for use as part of Aircraft Monitor and Control (AMAC) System A/A24B-4(V) to relay arming option and deliver mode signals from Control-Monitor DCK-143/A24B-4(V) and operating power from Transformer-Rectifier TEK-9/A24B-4(V) to a nuclear weapon, and weapon status signals from the weapon to the control-monitor.

6.2 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of the specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in DL 68A27E1, it is desired that their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) The Drawings of DL 68A27E1 - The drawings shall have precedence over other referenced specifications. Any deviation from the drawings shall be specifically approved by the procuring activity.

(4) Referenced Specifications - Any referenced specifications shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing (see 5.1).

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6.5 Associated Equipment - The equipment shall operate with the following associated equipment:

(1) Control-Monitor DCK-143/A24B-4(V)

(2) Transformer-Rectifier TEK-9/A24B-4(V)

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Naval Air Systems Command
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SPECIFICATION ANALYSIS SHEET

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Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-R-81766(AS) RELAY PANEL RYK-13/A24B:(V)

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426

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